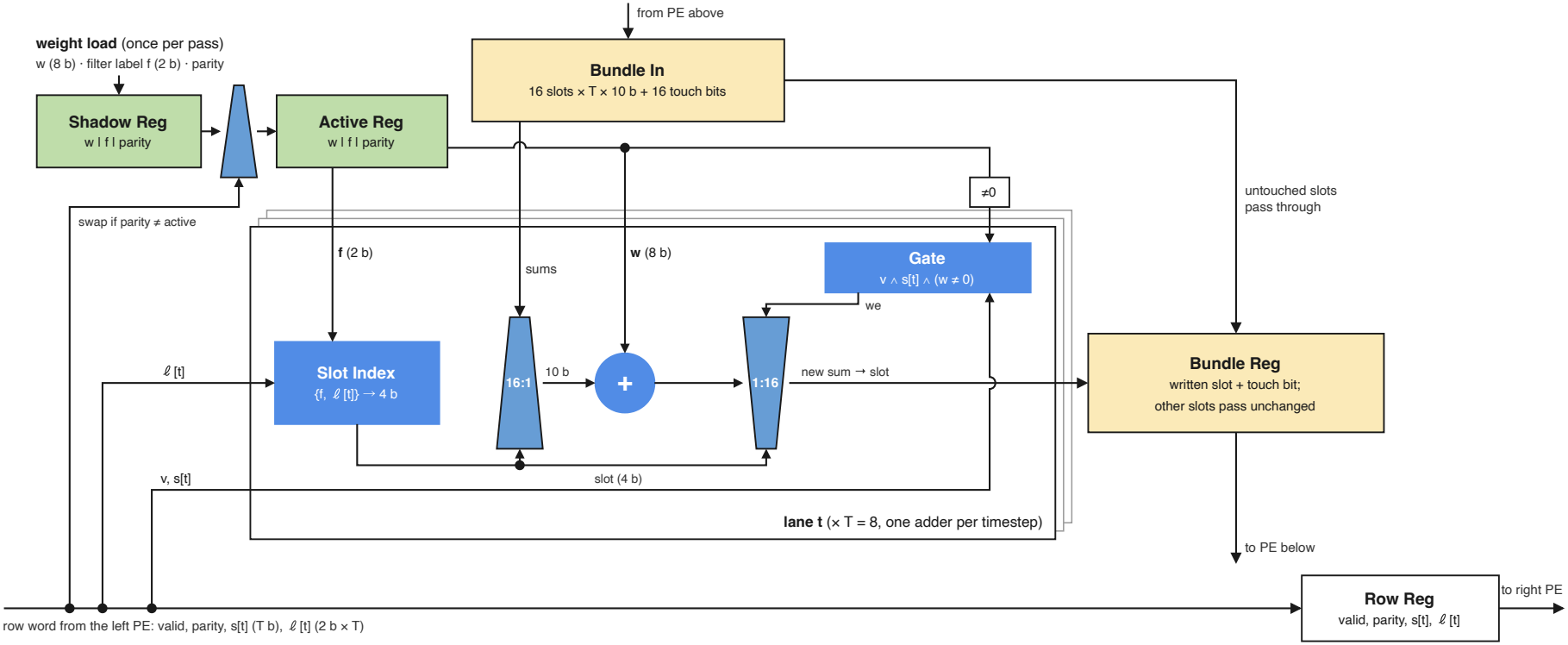
from PE above
weight load (once per pass)
w (8 b) · filter label f (2 b) · parity
Bundle In
16 slots × T × 10 b + 16 touch bits
Shadow Reg
w l f l parity
Active Reg
w l f l parity
≠0
untouched slots
pass through
swap if parity ≠ active
f (2 b)
sums
w (8 b)
Gate
v ∧ s[t] ∧ (w ≠ 0)
we
ℓ [t]
Slot Index
{f, ℓ [t]} → 4 b
16:1
10 b
+
1:16
new sum → slot
Bundle Reg
written slot + touch bit;
other slots pass unchanged
v, s[t]
slot (4 b)
lane t (× T = 8, one adder per timestep)
to PE below
Row Reg
valid, parity, s[t], ℓ [t]
to right PE
row word from the left PE: valid, parity, s[t] (T b), ℓ [t] (2 b × T)